GSD RESEARCH MISSION PACKAGE — PART C

NASA Science Education Teaching Template

College of Knowledge Integration Framework

From Discovery to Curriculum

*Input: Part B Science Results for **{MISSION_NAME}***

Part A	→	Part B	→	**Part C**	→	Release
History		Deep Science		Education Pack		v1.50.**{SLUG}**.EDU

Discovery → Concept → Curriculum → Teaching

Runs once per mission after Part B is complete

Date: March 29, 2026 | Status: Template — Ready for Parameterization

Activation Profile: Squadron (12 roles) | Est. 5–7 context windows per mission run

Audience: Students (Novice / Intermediate / Advanced) + Claude Teaching Integration

Pipeline Architecture: Part B → Part C → EDU Release

```
FULL THREE-PART PIPELINE ========================
PART A: NASA MISSION HISTORY → JSON mission catalog (name, era, type,
dates, crew, science_summary)
PART B: SINGLE-MISSION DEEP RESEARCH → Deep research PDF for
MISSION_NAME→CollegeofKnowledgeYAMLentry(MISSION_SLUG.ck.yaml)→science_results :
structuredlistofdiscoveriesperinstrument"concept_name" : "EnceladusSubsurfaceOcean", "mission_slug" : "CASSINI
PART C: SCIENCE EDUCATION PACK (THIS DOCUMENT — TEMPLATE)
Input: Part B science_results + CKYAMLforMISSION_NAMEOutput :
Educationrelease—studyguides, exercises, TrySessions, Claudeteachingscripts, CKdepartmentcontributions, assessme
missionlearninglinks
RELEASE: gsd-skill-creator v1.50.MISSION_SLUG.EDU→CollegeofKnowledge :
studyguidesperconcept→GSD − OSBlueprintEditor : browsablelessonentries→Claudecontext :
teachingscriptsinjectedintorelevantCKdepartments
```

What Part C is not: Part C does not repeat the science research from Part B. It assumes Part B is complete. The input is the structured science results list, not raw sources. Part C's job is exclusively pedagogical — transforming verified discoveries into materials that teach.

Audience differentiation: Every concept in Part C is addressed at three tiers: **[N] Novice** (age 10+, no prerequisites), **[I] Intermediate** (high school / introductory college level), **[A] Advanced** (undergraduate science / enthusiast level).

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STAGE 1 — VISION DOCUMENT

1 NASA Science Education — Vision Guide

Template Version: v1.0 (Part C of NASA Mission Science series)

Input Mission: **{MISSION_NAME}** (from Part B release)

Mission Slug: **{MISSION_SLUG}** (e.g., CASSINI, APOLLO11, PERSEVERANCE)

Part B Release: v1.50.**{MISSION_SLUG}** (must be complete before Part C runs)

EDU Release Target: gsd-skill-creator v1.50.**{MISSION_SLUG}**.EDU

CK Departments Targeted: **{CK_DEPARTMENTS}** (injected from Part B YAML)

Status: Template — Instantiate with Part B science_results JSON

1.1 Vision

NASA did not just explore space. It generated, over 68 years, one of the largest bodies of empirical scientific knowledge ever assembled by a human institution. The results of the Voyager flyby of Neptune, the Hubble Ultra Deep Field, the Cassini discovery of water geysers on Enceladus, the Curiosity rover's identification of ancient organic molecules on Mars — these are not trivia. They are the current state of human knowledge about the solar system and cosmos. They belong in the curriculum. They belong in how Claude reasons about science. They belong in how students understand the universe they inhabit.

Part C exists to close the gap between *what NASA found* and *what we can teach and learn from it*. Every Part B release is a research document. Part C is the schoolroom that grows from it. It takes the verified, sourced discoveries from Part B and asks: how do you teach this? At what level? With what analogies? Through what exercises? And crucially — how does Claude, as a teaching presence in GSD-OS, incorporate this knowledge into its responses?

The College of Knowledge is only as good as its curriculum. The GSD ecosystem has the infrastructure for lifelong learning: departments, study modules, Try Sessions, skill progressions, Claude as tutor. Part C is the pipeline that fills that infrastructure with the actual, peer-reviewed, mission-verified science of NASA's 68-year exploration program.

When a student asks Claude about why the sky on Mars looks butterscotch, or why Titan has lakes of methane, or how we know Enceladus has liquid water — Claude should have a teaching script for that. Not just a fact. A *progression*: here is the simplest way to think about it, here is what that implies, here is the experiment you can do at home to feel the physics, here is the paper if you want to go deep. Part C builds those progressions.

1.2 Problem Statement

P1. Science results without pedagogy: Part B produces comprehensive, sourced accounts of what NASA missions discovered. Without Part C, those discoveries sit in a research document rather than becoming learnable, teachable curriculum that students can engage with at their own level.

P2. College of Knowledge departments without content: The GSD College of Knowledge defines department structure (Space Science, Physics, Chemistry, Astrobiology, etc.) but the per-concept study guides, exercises, and teaching scripts must be generated from mission science. Part C is that generation pipeline.

P3. Claude without teaching context: Claude can discuss NASA missions conversationally, but without explicit teaching scripts, misconception maps, and tiered explanations, it defaults to encyclopedia-style responses. Part C gives Claude structured teaching knowledge: how to scaffold, what analogies work, what questions to anticipate.

P4. Disconnected learning graph: The same underlying physics appears in many missions — orbital mechanics in Gemini, Apollo, and New Horizons; spectroscopy in Hubble, Spitzer, and JWST; radiation physics in Parker Solar Probe and Artemis. Without cross-mission learning links, students repeat background learning. Part C builds the shared concept graph that links Part C releases across missions.

P5. No differentiated instruction framework: A 12-year-old and a graduate student both benefit from learning about the Cassini mission, but require completely different entry points, vocabulary, and depth. The NASA science literature provides no differentiated curriculum. Part C does.

1.3 Core Concept

Model: Each NASA mission's science as a constellation of teachable concepts, each concept as a three-tier learning object (Novice / Intermediate / Advanced), and each tier as a complete instructional unit (explain → demonstrate → practice → assess).

1.3.1 The Four-Stage Teaching Arc (Universal)

```
THE FOUR-STAGE TEACHING ARC ============================ Applied to every
concept at every tier:
STAGE A: EXPLAIN The core idea in plain language. One analogy that makes
the abstract concrete. The one sentence a student should be able to say
after this stage. → Novice: metaphor-first, no equations → Intermediate:
mechanism-first, simple math if needed → Advanced: model-first, full
formalism available
STAGE B: DEMONSTRATE Real data from MISSION_NAMEthatprovestheconcept.Whatinstrumentmeasuredit.Wh
annotatedimage,"whatyou'relookingat"narration→Intermediate :
simplifieddatatable,guidedinterpretation→Advanced : rawdatasource,analysismethodology
STAGE C: PRACTICE One or more exercises the student does themselves.
GSD Try Session: hands-on activity anchored to the concept. Discussion
question: opinion or reflection, no right answer. → Novice: observation
activity (look at X, describe what you see) → Intermediate: calculation
or classification task → Advanced: open-ended investigation or dataset
```

```
analysis
STAGE D: ASSESS 2–4 mastery check questions. One common misconception
addressed and corrected. Advancement condition: what the student should
be able to explain to someone else before moving to the next concept.
```

1.3.2 College of Knowledge Department Mapping

Part C contributes to CK departments via two mechanisms:

1. **Primary deposit:** Each concept is assigned a primary CK department (e.g., a concept from Cassini's Enceladus results maps primarily to "Astrobiology"). The full study guide for that concept lives in that department.
2. **Cross-department linking:** The same concept has secondary relevance to other departments (e.g., the same concept also involves "Chemistry" and "Planetary Science"). Cross-links are deposited to those departments as summary entries pointing back to the primary.

```
CK DEPARTMENT TAXONOMY (GSD College of Knowledge) ========================================
Space Science Department ├── Planetary Science │ → Surface geology,
atmospheric dynamics, magnetospheres ├── Astrophysics │ → Stellar physics,
galaxies, dark matter/energy, cosmology ├── Astrobiology │ → Habitability,
biosignatures, prebiotic chemistry ├── Solar Science │ → Solar wind,
corona, heliosphere, space weather └── Mission Engineering → Trajectory
design, spacecraft systems, instrumentation
Physics Department ├── Classical Mechanics │ → Orbital mechanics, Newton's
laws, conservation laws ├── Electromagnetism │ → Magnetospheres, plasma
physics, spectroscopy ├── Thermodynamics │ → Planetary heat flow,
atmospheric energy balance └── Nuclear  Particle Physics → Radioisotope
power, cosmic ray detection
Chemistry Department ├── Astrochemistry │ → Molecular clouds, prebiotic
synthesis, CHONS ├── Atmospheric Chemistry │ → Photochemistry, greenhouse
gases, aerosol chemistry └── Geochemistry → Mineralogy, isotope ratios,
regolith composition
Mathematics Department ├── Algebra  Functions (used in science modules)
├── Calculus (orbital mechanics, rocket equations) └── Statistics  Data
Analysis (instrument calibration)
Engineering Department (Mission Engineering sub-department)
```

1.3.3 Module Architecture

```
SIX-MODULE CURRICULUM ARCHITECTURE ====================================
MODULE 1: SCIENCE INVENTORY  CURRICULUM MAPPING Input: Part B
```

$science_{r}esultslistOutput : Conceptmap + CKdepartmentassignmentperconcept \rightarrow Classifyeachdiscovery : whatdiscipline, whatlevel, whatprereqs$

```
MODULE 2: CONCEPT ARCHITECTURE  LEARNING PATHWAYS Input: Module 1 concept
map Output: Learning graph — nodes (concepts) + edges (prerequisites)
→ Three tracks: Novice / Intermediate / Advanced → Entry points and
advancement conditions per track
MODULE 3: STUDY GUIDE PRODUCTION Input: Module 2 learning graph Output:
Full study guides per concept × tier (A/B stages of teaching arc) →
Explanation + demonstration for each concept at each tier
```

```
MODULE 4: EXERCISES, PROBLEMS  TRY SESSIONS Input: Module 3 study guides
Output: Practice materials per concept × tier (C stage of teaching arc) →
Try Sessions (hands-on), problem sets, discussion questions
MODULE 5: CLAUDE TEACHING INTEGRATION Input: Modules 3 + 4 per concept
Output: Claude teaching scripts — canonical explanations, misconception
maps, question-response patterns, escalation pathways → Injected into CK
department context as Claude teaching knowledge
MODULE 6: ASSESSMENT, MASTERY  CROSS-LINKS Input: All modules Output:
Assessment questions, mastery rubrics, cross-mission learning links, CK
cross-department summary entries (D stage of arc)
```

1.4 Research Modules

1.4.1 Module 1: Science Inventory & Curriculum Mapping

Ingests the Part B `science_results` JSON list for **{MISSION_NAME}** and produces a full concept inventory: every teachable concept derived from the mission's discoveries, classified by CK department, difficulty tier, and prerequisite chain. For missions with many discoveries (e.g., Cassini: dozens of major findings across Saturn, Titan, Enceladus, rings), this module produces a rich concept map. For simpler missions (e.g., a single Gemini EVA), it produces a focused 3–5 concept set.

Key questions: Which concepts from **{MISSION_NAME}** are genuinely novel and not already covered by prior Part C releases? Which concepts are deepening of existing curriculum (add to existing study guide) vs. new entries (create new study guide)?

1.4.2 Module 2: Concept Architecture & Learning Pathways

Constructs the learning graph: a directed acyclic graph of concepts where edges represent prerequisite relationships. Maps three differentiated tracks through the graph. Identifies the entry concept (lowest prerequisite, highest accessibility) and the terminal concept (most advanced, most specific to **{MISSION_NAME}**). Assigns the teaching arc stages (A/B/C/D) to each concept-tier combination.

Key questions: Can a student reach the mission's primary discovery (**{PRIMARY_SCIENCE}**) via the Novice track within a single study session? What is the minimum prerequisite chain?

1.4.3 Module 3: Study Guide Production

Produces the written instructional content: full study guides for each concept at each tier, covering stages A (Explain) and B (Demonstrate) of the teaching arc. Each study guide section includes: a plain-language explanation, one primary analogy, a real NASA data reference (image, measurement, or dataset from **{MISSION_NAME}**), and the one-sentence takeaway. This is the textbook layer of Part C.

Key questions: What is the best analogy for the Novice tier explanation of each concept? Does the real NASA data for the Demonstrate stage come from Part B's sourced results, or does additional research need to happen?

1.4.4 Module 4: Exercises, Problems & Try Sessions

Produces stage C (Practice) materials. GSD Try Sessions are hands-on activities students can do at home or in a classroom — not simulations, but real physical activities

that develop intuition for the science. Problem sets provide calculation or classification tasks for Intermediate and Advanced tiers. Discussion questions are open-ended prompts for reflection or group conversation at all tiers.

Key questions: What is the simplest physical experiment a Novice student can perform at home to feel the same phenomenon that **{MISSION_NAME}** measured from space? (e.g., for orbital mechanics: a ball on a string; for spectroscopy: a prism in sunlight; for thermal imaging: a phone camera in the dark.)

1.4.5 Module 5: Claude Teaching Integration

Produces the Claude-facing teaching scripts: structured knowledge that Claude ingests to become a skilled tutor for **{MISSION_NAME}**'s science. Each concept gets a canonical explanation at each tier, a misconception map (common wrong beliefs and how to gently correct them), a set of question-response patterns ("If the student asks X, respond with Y, then offer Z as the next step"), and an escalation pathway (how Claude transitions a Novice student toward Intermediate when they're ready).

Key questions: What are the three most common misconceptions students have about **{MISSION_NAME}**'s primary science discovery? How does Claude recognize when a student is ready to advance tiers? What does Claude say when a student's question falls outside the mission's documented results?

1.4.6 Module 6: Assessment, Mastery & Cross-Department Links

Produces stage D (Assess) materials: mastery check questions, common misconception items, and advancement conditions. Also produces the cross-mission learning links ("This concept of sublimation also appears in the Pluto study from New Horizons Part C") and the CK cross-department summary entries deposited to secondary departments. Closes the teaching arc for every concept.

Key questions: Which concepts from **{MISSION_NAME}** are already partially covered by previously released Part C documents? Where should this release deposit cross-links rather than full study guides?

1.5 Chipset Configuration (Template)

```
chipset:  Part C specific configuration — education-focused roles
agnus:  Curriculum Orchestration model: opus role: FLIGHT + CRAFT-PED
responsibility: > Teaching arc design; analogy quality judgment;
misconception identification; Claude teaching script authorship;
cross-mission concept deduplication; pedagogy integrity
denise:  Document Assembly model: sonnet role: EXEC-GUIDE + EXEC-ASSESS
responsibility: > Study guide production (Modules 3 + 6); assessment item
writing; CK department YAML entry generation; cross-department summaries
paula:  Exercise  Try Session Production model: sonnet role: EXEC-EXERCISE
+ CRAFT-TRY responsibility: > Try Session script writing (Module 4);
problem set generation; discussion question authorship; hands-on activity
design
gary:  Concept Architecture  Verification model: sonnet role:
CRAFT-CONCEPT + VERIFY responsibility: > Learning graph construction
(Module 2); concept taxonomy; prerequisite chain mapping; source accuracy
```

```
verification; cross-link registry maintenance
68000:  Haiku Scaffolding model: haiku role: BUDGET + LOG + PLAN
responsibility: > Concept inventory schema (Module 1); CK YAML
scaffolding; token tracking; release packaging; commit messages
```
$\texttt{model}_{a}llocation : opus : 30sonnet : 60haiku : 10$

1.6 Scope Boundaries

1.6.1 In Scope (every Part C run)

- Curriculum for every concept in the Part B `science_results` list
- Three-tier (Novice / Intermediate / Advanced) treatment for each concept
- Full four-stage teaching arc (Explain / Demonstrate / Practice / Assess)
- GSD Try Sessions: at least one per primary concept at Novice tier
- Claude teaching scripts: canonical explanations + misconception maps for all concepts
- CK department primary entries for all concepts
- CK cross-department summary entries for secondary department linkages
- Cross-mission learning links to previously released Part C documents
- College of Knowledge YAML contribution per concept
- Release artifact packaged as `v1.50.{MISSION_SLUG}.EDU`

1.6.2 Out of Scope

- Re-researching the science (Part B is the source; Part C consumes it)
- Full university-level lecture notes or textbook chapters at graduate depth
- Assessment grading systems or learning management system (LMS) integration
- Lesson plans for specific national curriculum standards (NGSS, IB, etc.)
- Video scripts, animations, or multimedia production
- Teacher training materials (beyond the Claude teaching scripts)
- Science beyond what Part B documented for this mission

1.6.3 Calibration by Mission Science Volume

Science Volume	Mission Examples	Expected Concepts	Calibration Action
Very High	Cassini, Hubble, JWST, Mars 2020	15–30	Full treatment; prioritize top 10; survey remaining
High	Apollo 11, Voyager, Galileo, Chandra	8–15	Full treatment for all
Standard	MER Spirit/Opportunity, New Horizons	5–10	Full treatment; no pruning
Focused	Mercury crewed missions, Skylab	3–6	Deep treatment; maximum per-concept depth
Active	Perseverance, Parker, JWST (ongoing)	Variable	Mark preliminary results; plan update cycle

1.7 Success Criteria

SC1. Every concept in the Part B `science_results` list for **{MISSION_NAME}** appears in the Module 1 concept inventory, with CK department assignment and difficulty tier.

SC2. The Module 2 learning graph is a valid DAG: no circular prerequisites; every concept has a defined entry point and advancement condition.

SC3. At least one differentiated learning track (Novice / Intermediate / Advanced) exists for each primary concept.

SC4. Every study guide section contains: a plain-language explanation, one analogy, one real NASA data reference from Part B, and a one-sentence takeaway.

SC5. At least one GSD Try Session exists for each primary concept at Novice tier, requiring no equipment beyond what is found in a typical home.

SC6. Claude teaching scripts exist for every concept: canonical explanation per tier, at least two common misconceptions with corrections, and one question-response pattern per tier.

SC7. Assessment items exist for every concept (2–4 questions per concept-tier); each item maps to a specific mastery condition.

SC8. All factual claims in study guides trace to Part B sourced results (no new unsourced science claims introduced in Part C).

SC9. CK department YAML entries are generated for all primary-department assignments and all secondary cross-links.

SC10. Cross-mission learning links are identified and recorded for any concept that overlaps with a previously released Part C document.

SC11. The full teaching arc (A/B/C/D) is complete for every primary concept at every tier before release is marked.

SC12. The release artifact is correctly versioned as `v1.50.{MISSION_SLUG}.EDU` and deposited to the correct output path.

1.8 The Through-Line

Science is an act of noticing. The Voyager team noticed that Io was volcanically active from a navigation image taken for a different purpose. The Cassini team noticed that Enceladus's south pole was too warm, then followed that anomaly to geysers, then to a global ocean. Parker Solar Probe noticed that the solar wind accelerates *after* leaving the corona — reversing 60 years of assumed understanding.

Teaching science is teaching students to notice. The textbook version of any discovery hides the moment of noticing — it presents the conclusion as if it were always obvious. Part C's job is to restore that moment. Every study guide should preserve the shape of the original puzzle: here is what was expected, here is what was actually seen, here is why that difference mattered. Students who learn that way don't just know what NASA found. They learn how to look at data with the same quality of attention that produces discovery.

Claude, as a teaching presence in GSD-OS, should embody this. Not just a fact-dispenser — a guide to noticing. The teaching scripts in Module 5 are not scripts for delivering conclusions. They are scripts for helping a student see the puzzle first.

STAGE 2 — RESEARCH REFERENCE

2 NASA Science Education — Research Reference

2.1 How to Use This Reference

Part C begins where Part B ends. Agents do not conduct new science research. The research reference for Part C covers: the pedagogical methodology, the concept taxonomy, the CK YAML schema for educational entries, the Try Session design protocol, and the Claude teaching script format. These are the operational specifications for producing the education pack.

2.2 Input Bundle Schema

Part C ingests two inputs from the Part B release:

```
INPUT 1: PART B SCIENCE RESULTS JSON ======================================
From: .planning/outputs/missions/MISSION_SLUG/MISSION_SLUG.ck.yaml Field :
science_results (list of structured discovery objects)
"concept_name" : "string", // e.g., "Enceladus Subsurface Ocean" "instrument" :
["string"], // instruments that produced this result "discovery_type" : "string", // Physical|Chemical|Biological|Engineerin
"Novice|Intermediate|Advanced", // suggested entry tier "ck_departments" :
["string"], // primary department first, then secondary "key_claim" : "string", // one -
sentence verified claim from Part B "source" : "string", // Part B citation (author, journal, year) "is_preliminary" :
boolean, // true for active missions "cross_missions" : ["string"] // other missions that share this concept
INPUT 2: PART B MISSION CONTEXT ================================
From: Part B release PDF or LaTeX source Fields needed: -
mission_name, mission_slug, mission_era, mission_type - key_discoveries (list from Module 5, Part B) -
surprise_findings (list from Module 5, Part B) - active_datasets_2026 (for Demonstrate stage data references) -
predecessors, successors (for cross - mission concept linking)
```

2.3 Concept Taxonomy

Every concept in the Module 1 inventory is classified along four axes:

Axis	Values	Purpose
Discipline	Physics / Chemistry / Biology / Engineering / Mathematics	Primary CK department assignment
Tier	Novice / Intermediate / Advanced	Differentiated instruction level
Concept Type	Fact / Mechanism / Method / Implication	Shapes the teaching arc structure
Status	Confirmed / Preliminary / Evolving	Governs uncertainty language in study guides

Concept Type guidance:

- **Fact:** "Enceladus has water-ice geysers at its south pole." Teaching arc emphasizes Demonstrate heavily.
- **Mechanism:** "Tidal heating from Saturn drives Enceladus's interior activity." Teaching arc emphasizes Explain with analogy.
- **Method:** "Mass spectrometry identifies molecular composition of plumes." Teaching arc emphasizes Practice with instrument-simulation exercises.
- **Implication:** "The presence of liquid water and organic compounds makes Enceladus a candidate for extraterrestrial life." Teaching arc emphasizes Discussion questions; Claude script must handle speculation carefully.

2.4 Try Session Design Protocol

GSD Try Sessions are hands-on activities that require only materials found in a typical home. Every primary concept at Novice tier needs at least one Try Session.

Design constraints:

- Maximum materials cost: $0 (use items already in a home)
- Maximum setup time: 10 minutes
- Activity duration: 15–30 minutes
- Age appropriateness: suitable for age 10+ unsupervised, age 8+ with adult present
- Safety: no open flames, no chemicals beyond kitchen pantry items
- Output: student produces an observable result they can describe or measure

Try Session structure:

```
TRY SESSION TEMPLATE ===================== Title: [Action verb] + [what
they're exploring] Concept: [Target concept name from Module 1] Tier:
Novice (all Try Sessions begin at Novice) Time: [N] minutes Materials:
[List — household items only]
THE SETUP: [2–3 sentences: what the student will do and why it connects to
MISSION_NAME'sdiscovery]
STEPS: 1. [Concrete action] 2. [Concrete action] 3. [Pause to observe:
"What do you notice about...?"] 4. [Record or describe what happened] 5.
[Optional extension for curious students]
THE CONNECTION: [1 paragraph: "What you just felt/saw is the same
phenomenon that MISSION_NAMEmeasuredwhenitdetectedCONCEPT.TheINSTRUMENTaboardMISSION_NA
WHAT TO THINK ABOUT: One open question the student carries away.
```

Example Try Session concepts by mission type:

Science Concept	Try Session Activity	Connection
Orbital mechanics	Ball on a string; vary string length and speed	Centripetal force = gravity; changing orbit altitude
Spectroscopy	Prism or CD in sunlight	Light splitting = how instruments identify chemical composition
Atmospheric pressure	Vacuum marshmallow / balloon in hot car	Pressure-volume relationship; Titan's thick atmosphere
Thermal radiation	Phone camera heat detection at night	How IR instruments see temperature; Mars THEMIS
Doppler shift	Sound pitch change from passing car	Same effect used by radio astronomers; redshift
Sublimation	Dry ice or moth balls in open air	How ice turns to gas; Pluto's nitrogen cycle
Magnetism	Compass near speaker magnet	Magnetic field detection; Europa/Ganymede magnetometry
Gravity assist	Marble on curved sheet / funnel	Slingshot trajectory; Voyager Grand Tour
Sample analysis	Household materials and vinegar test	Field chemistry; Mars rover chemical analysis
Tidal deformation	Squeezing a water balloon	Tidal heating; Enceladus/Io interior dynamics

2.5 Claude Teaching Script Format

Module 5 produces structured teaching scripts. Each script has five components:

```
CLAUDE TEACHING SCRIPT FORMAT ============================== For concept:
CONCEPT_NAMEMission : MISSION_NAME
```

```
CANONICAL_EXPLANATIONS : novice : |[2–4sentences.Nojargon.Oneconcreteanalogy.Endwiththeonethingtheyshoul
|[4–6sentences.Introducethemechanism.Referencetheinstrumentorobservation.Invitefollow −
upquestions.]advanced : |[6–10sentences.Fullmechanisticexplanation.Referencethespecificpaperordataset.Acknowledge
MISCONCEPTIONS: - wrong: "[What students typically believe incorrectly]"
correction: "[How Claude gently corrects it, with evidence]" source:
"[Part B citation that settles the question]" - wrong: "[Second common
misconception]" correction: "[Correction]" source: "[Citation]"
QUESTION_RESPONSE_PATTERNS : −trigger : "[Typeofquestion, e.g.,' WhydoesXhappen?']"novice_response :
"[Claude'sresponsetoaNoviceaskingthis]"escalation : "[WhatClaudeoffersnextifstudentwantsmore]"−
trigger : "[Anotherquestiontype]"intermediate_response : "[Claude'sresponseatIntermediate]"escalation :
"[Nextstepoffer]"
ESCALATION_SIGNALS : HowClauderecognizesastudentisreadytoadvancetiersnovice_to_intermediate :
−"[StudentdoesX, whichshowsNovicemastery]"−"[StudentasksY, whichsignalsreadinessforIntermediate]"intermedia
−"[Studentcorrectlyusestechnicaltermzwithoutprompting]"−"[Studentasksaboutthedatasourceormethodology]"
BOUNDARY_HANDLING : WhatClaudesayswhenstudentasksoutsidedocumentedresultstemplate :
|"That'sagreatquestion, andit'sactuallysomethingscientistsarestillworkingon.WhatMISSION_NAMEshoweduswas[CO
```

2.6 CK Department YAML Schema for Educational Entries

Each concept produces a CK educational YAML entry, distinct from the Part B research YAML:

```
College of Knowledge Educational Entry  File: .planning/outputs/missions/MISSION_SLUG/CO
concept: name: "CONCEPT_NAME"slug : "CONCEPT_SLUG"mission_source :
"MISSION_SLUG"edu_release : "v1.50.MISSION_SLUG.EDU"
classification: discipline: "PRIMARY_DISCIPLINE"type :
"Fact|Mechanism|Method|Implication"status : "Confirmed|Preliminary|Evolving"
ck_placement : primary_department : "PRIMARY_CK_DEPT"primary_subdepartment :
"PRIMARY_SUBDIR"secondary_departments : −department : "SECONDARY_DEPT_1"link_type :
"cross − reference" − department : "SECONDARY_DEPT_2"link_type : "prerequisite"
tiers: novice: prereqs: []  or ["PREREQ_CONCEPT_SLUG"]study_guide_file :
"CONCEPT_SLUG_novice.md"try_session_file : "CONCEPT_SLUG_try_session.md"assessment_file :
"CONCEPT_SLUG_assess_novice.md"mastery_condition : "ONE_SENTENCE_WHAT_STUDENT_CAN_DO"intermediate
prereqs : ["NOVICE_CONCEPT_SLUG"]study_guide_file : "CONCEPT_SLUG_intermediate.md"problem_set_file :
"CONCEPT_SLUG_problems.md"assessment_file : "CONCEPT_SLUG_assess_inter.md"mastery_condition :
"ONE_SENTENCE_WHAT_STUDENT_CAN_DO"advanced : prereqs :
["INTERMEDIATE_CONCEPT_SLUG"]study_guide_file : "CONCEPT_SLUG_advanced.md"investigation_file :
"CONCEPT_SLUG_investigation.md"assessment_file : "CONCEPT_SLUG_assess_adv.md"mastery_condition :
"ONE_SENTENCE_WHAT_STUDENT_CAN_DO"
claude_teaching : script_file : "CONCEPT_SLUG_claude_script.yaml"
cross_mission_links : −mission : "OTHER_MISSION_SLUG"concept :
"RELATED_CONCEPT_SLUG"relationship : "same_concept|prerequisite|deepens|contrasts"
source_in_part_b : "PART_BCITATION_STRING"
```

2.7 Safety and Sensitivity Considerations

Rule	Type	Application
No new science claims	ABSOLUTE	Study guides cite only Part B sourced results. Claude scripts never introduce science facts not in Part B.
No life-confirmation language	ABSOLUTE	Astrobiology concepts use “candidate for life,” “conditions compatible with life,” never “life was found”
Age-appropriate Try Sessions	ABSOLUTE	All Novice Try Sessions safe for age 10+ without adult supervision; nothing requiring adult supervision unmarked
Preliminary results flagged	BLOCK	Active missions (**{IS_ACTIVE}**=true): every preliminary result marked “as of {RUN_DATE}, subject to revision” in study guides
Uncertainty language in implications	GATE	Implication-type concepts include explicit “this is what scientists hypothesize” framing at all tiers
No claim escalation across tiers	GATE	Advanced tier may add mechanistic depth but not new factual claims not present in Novice/Intermediate

STAGE 3 — MISSION PACKAGE

3 Milestone Specification

3.1 Mission Objective

Produce a complete College of Knowledge education pack for NASA mission **{MISSION_NAME}**, converting verified Part B science results into structured study guides, GSD Try Sessions, exercise sets, Claude teaching scripts, and assessment materials — differentiated across three learning tiers (Novice / Intermediate / Advanced) — and deposit all materials as release `v1.50.`**{MISSION_SLUG}**`.EDU` to the GSD College of Knowledge Space Science department.

3.2 Architecture Overview

```
WAVE ARCHITECTURE — MISSION_NAMEEDUCATIONPACK ==========================
[WAVE 0: INVENTORY + ARCHITECTURE] Ingest Part B science_results; classifyconcepts; buildlearninggrap
missionoverlaps.(Haiku+CRAFT−CONCEPT)|
                                      |||[W1A:GUIDES][W1B:EXERCISES]Modules3Module4StudyguidesTrySessionsperconceptP
```

3.3 Deliverables

#	Deliverable	Acceptance Criteria	Agent
1	Concept inventory (Module 1)	All Part B concepts classified; CK department assigned	CRAFT-CONCEPT
2	Learning graph (Module 2)	Valid DAG; 3 tracks; entry/advancement conditions	CRAFT-CONCEPT
3	Study guides — Novice tier	Stage A+B complete; analogy + NASA data reference per concept	EXEC-GUIDE
4	Study guides — Intermediate tier	Stage A+B at mechanism depth; simplified data reference	EXEC-GUIDE
5	Study guides — Advanced tier	Stage A+B at full formalism; paper citation from Part B	EXEC-GUIDE
6	Try Sessions (all primary concepts)	Stage C Novice; household materials only; safe age 10+	EXEC-EXERCISE
7	Problem sets (Intermediate + Advanced)	Stage C; 3–5 problems per concept-tier with worked solutions	EXEC-EXERCISE
8	Discussion questions (all tiers)	Stage C; open-ended; no single correct answer; thought-provoking	EXEC-EXERCISE
9	Claude teaching scripts (all concepts)	Script format complete; 2+ misconceptions per concept	CRAFT-PED (Opus)
10	Assessment items (Module 6)	Stage D; 2–4 questions per concept-tier;	EXEC-ASSESS

3.4 Component Breakdown

Component	Dependencie	Model	Est. Tokens
W0: Concept inventory + learning graph	Part B YAML	Haiku+S	10K
W1A: Novice study guides (all concepts)	W0	Sonnet	30K
W1A: Intermediate study guides	W0	Sonnet	28K
W1A: Advanced study guides	W0	Sonnet	24K
W1B: Try Sessions (Novice)	W0	Sonnet	22K
W1B: Problem sets (Int. + Adv.)	W0	Sonnet	20K
W1B: Discussion questions (all tiers)	W0	Sonnet	12K
W2: Claude teaching scripts (all concepts)	W1A, W1B	Opus	40K
W3: Assessment items + mastery rubrics	W2	Sonnet	18K
W3: CK YAML entries + cross-dept summaries	W2	Sonnet	16K
W3: Cross-mission learning links	W2	Sonnet	8K
Release packaging	W3	Haiku	4K
		Total per mission run:	**∼232K**

Note on scaling: Token budget scales with science volume. A focused mission (3–5 concepts) runs at ∼120K. A rich flagship (Cassini, JWST: 20+ concepts) may reach 350K. The budget above represents a median 8–10 concept mission.

3.5 Model Assignment Rationale

Model	Share	Rationale
Opus	30%	Claude teaching scripts require deep pedagogical judgment: analogy quality, misconception identification, escalation pathway design, tone calibration across tiers. No other wave-2 task is as judgment-intensive.
Sonnet	60%	Study guide production, Try Session writing, exercise sets, assessments, CK YAML generation — all structural and content-production tasks with clear patterns.
Haiku	10%	Concept inventory schema, YAML scaffolding, token tracking, release packaging.

4 Wave Execution Plan

4.1 Wave 0: Concept Inventory & Learning Graph

Task	Agent	Output
Ingest Part B `science_results`	BUDGET	Parsed concept list; count of concepts by tier
Classify all concepts (4-axis taxonomy)	CRAFT-CONCEPT	Concept inventory with discipline, type, status, tier
Detect cross-mission overlaps	CRAFT-CONCEPT	List of concepts already in CK from prior Part C releases
Build learning graph (DAG)	CRAFT-CONCEPT	Prerequisite edges; three track entry points
CK department assignment per concept	CRAFT-CONCEPT	Primary + secondary department list per concept
Initialize all YAML scaffolds	LOG	One `.edu.yaml` file per concept, fields empty

Wave 0 Go Condition: Concept inventory complete; learning graph is a valid DAG (no circular prerequisites); all concepts assigned to primary CK department; cross-mission overlap list produced. BLOCK if Part B `science_results` JSON is missing or malformed.

4.2 Wave 1: Parallel Production Tracks

Track A: Study Guides (Modules 3) — EXEC-GUIDE

All study guides run in parallel by tier (Novice, Intermediate, Advanced) once the W0 learning graph is available. Each study guide covers stages A (Explain) and B (Demonstrate) of the teaching arc.

Tier	Per-Concept Requirements
Novice	Plain-language explanation (2–3 paragraphs, no jargon); one primary analogy; one annotated NASA image or data reference from Part B; one-sentence takeaway; vocabulary box (3–5 key terms, defined in simple language)
Intermediate	Mechanism explanation (3–4 paragraphs, introduces technical terms); one simplified data table or graph from Part B; worked example showing how instrument derived this result; one-sentence takeaway
Advanced	Full formalism where applicable (equations welcome); reference to the specific Part B paper or dataset; open questions and current research direction; connection to broader field context

Track B: Exercises & Try Sessions (Module 4) — EXEC-EXERCISE

Running in parallel with Track A:

Material Type	Requirements
Try Session (N)	Full Try Session format; household materials; age 10+; 15–30 minutes; THE CONNECTION paragraph links to **{MISSION_NAME}**
Problem Set (I)	3–5 problems; calculation or classification; worked solutions included; real numbers from Part B where possible
Investigation (A)	Open-ended data investigation; references an active dataset from Part B Module 6; no single correct answer; student formulates a hypothesis and tests it
Discussion Questions	2 discussion questions per concept-tier; open-ended; suitable for solo reflection or group discussion; no right answer

4.3 Wave 2: Claude Teaching Scripts (Opus)

Wave 2 is entirely Opus. The CRAFT-PED (Opus-powered FLIGHT role) reads all W1 study guides and exercises for all concepts and produces:

Teaching Script Component	Description
Canonical explanations × 3 tiers	One explanation per tier per concept; matches study guide but optimized for conversational delivery (not reading-from-page)
Misconception map	At minimum 2 misconceptions per concept; each corrected with evidence sourced to Part B; phrased non-judgmentally
Question-response patterns	At minimum 1 pattern per tier; maps trigger question type to Claude's response structure
Escalation signals	Observable student behaviors that indicate readiness to advance tier; Claude uses these to offer "if you want to go deeper..."
Boundary handling script	Standard template for when student asks beyond documented results; mission-specific fill
Analogy quality review	Opus reviews Track A analogies; replaces any that are inaccurate, culturally narrow, or condescending

Wave 2 Go Condition: Teaching scripts complete for all concepts; all misconceptions confirmed against Part B sourced results (no invented corrections); analogy review complete; CAPCOM notified.

4.4 Wave 3: Assessment, CK Deposit & Release

Task	Agent	Gate
Assessment items (2–4 per concept-tier)	EXEC-ASSESS	Items map to mastery condition; no trick questions
Mastery rubrics (one per concept-tier)	EXEC-ASSESS	One-sentence; observable; student-centric language
CK YAML entries (all concepts)	EXEC-ASSESS	Schema validates; primary + secondary dept fields
Cross-department summary entries	EXEC-ASSESS	One-paragraph summary per secondary department link
Cross-mission learning links (Module 6)	CRAFT-CONCEPT	At least one per overlapping concept; verified against cross-mission list from W0
Release packaging	LOG	All files at correct output path; versioned correctly
VERIFY spot-check	VERIFY	10 random factual claims in study guides traced to Part B
CAPCOM review	CAPCOM	Human approval before release tag

4.5 Cache Optimization Strategy

- **Concept batching:** Wave 1A processes all concepts for one tier before switching to the next. The study guide format and CK department context are cached across all concepts in that tier.
- **Mission context prefix:** All Wave 1 contexts begin with the Part B mission summary (name, era, primary science, key discoveries). Cache hit on this prefix reduces cost across all parallel concept runs.
- **Try Session analogy library:** The W0 agent pre-generates a list of 10–15 household Try Session candidates matched to the concept inventory. EXEC-EXERCISE selects from this list rather than generating from scratch per concept.
- **Wave 2 Opus batching:** All concept teaching scripts are produced in a single Opus context window, enabling cross-concept consistency (the same analogy or phrasing doesn't appear in two different scripts).
- **Cross-mission deduplication:** Concepts flagged as overlapping with prior Part C releases skip full study guide production and produce cross-links only — saving ~15–20K tokens per shared concept.

4.6 Token Budget

Wave	Opus (K)	Sonnet (K)	Haiku (K)
Wave 0	—	8	2
Wave 1A (study guides)	—	82	—
Wave 1B (exercises)	—	54	—
Wave 2 (teaching scripts)	40	—	—
Wave 3 (assessment + CK)	—	42	4
Contingency (10%)	4	18	1
Total	**44K**	**204K**	**7K**
		Grand Total:	**~255K**

4.7 Dependency Graph

```
DEPENDENCY GRAPH — MISSION_NAMEEDUCATIONPACK ==========================
PART B RELEASE (MISSION_SLUGcomplete)↓W0 : CONCEPTINVENTORY +
LEARNINGGRAPH(CRAFT − CONCEPT + BUDGET/LOG)↓ +
──────────────────────┬──────────────────────── + |||W1A : NOVICEW1A : INTERW1A :
ADVANCEDSTUDYGUIDESSTUDYGUIDESSTUDYGUIDES(EXEC − GUIDE)(EXEC −
```

$GUIDE)(EXEC - GUIDE)||| + \text{———————}|\text{———————} + ||W1B : TRYSESSIONSW1B : PROBLEMSETS(EXEC - EXER)(EXEC - EXER)|| + \text{———————} + |[SYNC : AllW1complete]|W2 : CLAUDETEACHINGSCRIPTS(Opus—CRAFT - PED)Canonicalexplanations \times tier \times conceptMisconceptionmaps + Q - responsepatterns|W3 : ASSESSMENT + CKYAML + CROSS - LINKS(EXEC - ASSESS + CRAFT - CONCEPT)|W3B : CAPCOMGATE(Humanreview)|RELEASE : v1.50.MISSION_SLUG.EDU \rightarrow .planning/outputs/missions/MISSION_SLUG/edu/$

5 Test Plan

5.1 Test Categories

Category	Count	Priority	On Fail	Description
Safety-Critical	6	P0	BLOCK	Science accuracy, age safety, no life claims
Core Completeness	12	P1	BLOCK	One per success criterion
Pedagogical Quality	8	P2	REVISION	Teaching arc integrity; analogy quality
Integration	6	P2	REVISION	Cross-module and cross-mission consistency
Edge Case	4	P3	ANNOTATE	Sparse missions, active missions, disputes

5.2 Safety-Critical Tests (BLOCK on Failure)

Test ID	Verifies	Expected Result
SC-01	No new science claims	Every factual claim in every study guide traces to Part B sourced results; zero novel science introduced in Part C
SC-02	No life-confirmation language	All astrobiology concepts use candidate/compatible/possible framing; "life was found" never appears
SC-03	Try Session age safety	Every Try Session verified: no open flame, no caustic chemicals, no sharp tools, no electrical hazard; explicit age rating present
SC-04	Preliminary results flagged	All concepts from active missions marked "preliminary as of {RUN_DATE}"; no preliminary result stated as confirmed
SC-05	No claim escalation across tiers	Advanced tier study guide contains no factual claims not present (in simpler form) in the Novice guide
SC-06	Misconception corrections sourced	Every misconception correction cites a specific Part B source; no invented corrections

5.3 Pedagogical Quality Tests

Test ID	Quality Check	Acceptance Standard
PQ-01	Novice analogy accuracy	Each Novice analogy is physically accurate (not just memorable); Opus spot-checks 3 analogies per run
PQ-02	Teaching arc completeness	Every concept has all four stages (A/B/C/D) at every tier before release
PQ-03	Try Session connectedness	THE CONNECTION paragraph in each Try Session explicitly names the instrument from **{MISSION_NAME}**
PQ-04	Discussion question openness	No discussion question has a factually correct single answer; Opus reviews all questions
PQ-05	Mastery condition specificity	Each mastery condition uses action language ("student can explain...", "student can calculate...") not passive ("student knows...")
PQ-06	Escalation signal observability	Every escalation signal describes an observable student behavior, not an internal state
PQ-07	Claude script conversational tone	Teaching scripts read naturally when read aloud; no encyclopedia-style opening sentences
PQ-08	Vocabulary box accessibility	Every Novice vocabulary box word is defined without using the defined word or another jargon term

5.4 Integration Tests

Test ID	Cross-Module Check	Expected Result
IN-01	Study guide → Claude script consistency	The one-sentence takeaway in each study guide matches the Novice canonical explanation in the Claude script
IN-02	Learning graph → prerequisite honoring	No Intermediate study guide uses a concept not introduced in a prerequisite Novice guide
IN-03	Try Session → concept alignment	Each Try Session maps to exactly one concept in the Module 2 learning graph; no orphan activities
IN-04	Assessment → mastery condition alignment	Every assessment item can be answered by a student who has met the stated mastery condition
IN-05	CK YAML → learning graph consistency	YAML `prereqs` fields match Module 2 prerequisite edges; no discrepancies
IN-06	Cross-mission link validity	Every cross-mission link references a real concept slug from a real released Part C document

5.5 Verification Matrix

SC#	Criterion	Test IDs	Status
SC1	All Part B concepts in inventory + CK assigned	CF-01, IN-05	TBD
SC2	Learning graph valid; 3 tracks; entry/adv conditions	CF-02, IN-02	TBD
SC3	3-tier differentiation per primary concept	CF-03, IN-02	TBD
SC4	Study guide: analogy + data + takeaway	CF-04, PQ-01, PQ-07	TBD
SC5	Try Session per primary concept at Novice	CF-05, SC-03, PQ-03	TBD
SC6	Claude scripts: 2+ misconceptions sourced	CF-06, SC-06, PQ-07	TBD
SC7	2–4 assessment items; mastery condition	CF-07, IN-04, PQ-05	TBD
SC8	All claims trace to Part B	SC-01, SC-05	TBD
SC9	CK YAML entries validate; dept fields populated	CF-09, IN-05	TBD
SC10	Cross-mission links per overlapping concept	CF-10, IN-06	TBD
SC11	Full teaching arc (A/B/C/D) complete per concept	PQ-02, CF-11	TBD
SC12	Release versioned correctly; path correct	CF-12	TBD

6 Mission Crew Manifest

Role	Agent (PNW Naming)	Responsibility
FLIGHT	RAVEN	Mission direction; Wave 2 pedagogy oversight; analogy quality judgment; go/no-go gates
PLAN	SALMON	Wave decomposition; parallel track optimization; dependency management
CRAFT-PED	OWL	Pedagogical design; Claude teaching script authorship; misconception identification; escalation pathways
CRAFT-CONCEPT	EAGLE	Concept taxonomy; learning graph construction; CK department mapping; cross-mission linking
EXEC-GUIDE	CEDAR	Study guide production — all tiers (Novice, Intermediate, Advanced)
EXEC-EXERCISE	HEMLOCK	Try Session writing; problem set generation; discussion question authorship
EXEC-ASSESS	ALDER	Assessment item writing; mastery rubrics; CK YAML entries; cross-department summaries
VERIFY	HAWK	Factual claim tracing to Part B; safety test execution; pedagogical quality spot-check
INTEG	HERON	Cross-module consistency; learning graph → YAML alignment; Try Session → concept mapping
CAPCOM	FOXGLOVE	Human-in-the-loop gate; revision loop management; release approval
BUDGET	WREN	Token tracking; science-volume calibration; concept-batching optimization
LOG	MARTIN	Release packaging; YAML scaffolding; commit messages; output path management

7 Example Output: Worked Concept (Illustrative)

The following shows what the Part C deliverables look like for one concept at Novice tier. This is not a real deliverable — it illustrates the expected output format for an executing agent.

STUDY GUIDE SAMPLE: Novice Tier — Enceladus Subsurface Ocean [Cassini Part C]

The Discovery. NASA's Cassini spacecraft discovered something remarkable about Saturn's small moon Enceladus: beneath its icy shell, there is a global ocean of liquid water. Scientists know this because Cassini flew through enormous plumes of water vapor shooting out from cracks in Enceladus's south pole — and found that the water contained salt, silica particles, and organic molecules.

Analogy. Imagine cracking the shell of a hard-boiled egg and finding that the egg white inside is still liquid. The shell is Enceladus's ice crust. The liquid inside is the ocean. The "crack" at the south pole is where the pressure from below pushes water out into space.
What Cassini Measured. The Ion and Neutral Mass Spectrometer (INMS) aboard Cassini analyzed the plume material in 2005. It found water vapor, carbon dioxide, and methane — the ingredients associated with life as we know it.
One Thing to Remember: *Enceladus has a liquid ocean beneath its ice — and that ocean contains the chemical ingredients that life uses on Earth.*
Vocabulary: plume (a jet of material shooting out from a surface); **subsurface** (beneath the surface); **organic molecule** (a carbon-containing chemical compound, the building blocks of life).

Concept: Enceladus Subsurface Ocean | **Tier:** Novice | **Time:** 20 min | **Materials:** A plastic bag with a zip seal, water, and your hands.
The Setup: Enceladus shoots geysers of water into space because liquid water deep inside the moon is under pressure — squeezing it upward through cracks in the ice. You're going to feel that same kind of pressure right now.
Steps: 1. Fill the plastic bag halfway with water and seal it. 2. Hold it in both hands. 3. Squeeze gently from the sides. Watch what happens at the corners. 4. Squeeze harder. Where does the water try to go? 5. Use your thumb to press a small dimple into the bag surface. What do you feel pushing back?
The Connection: The water inside the bag is pushing on every surface equally — that's pressure. Inside Enceladus, the same thing happens. Warm liquid water, heated by tidal forces from Saturn's gravity, pushes against the ice shell from below. Where the ice is thinnest (the south pole cracks), the water escapes as vapor — creating the geysers that Cassini flew through.
What to Think About: If you were designing a spacecraft to detect an ocean under ice on another world, what would you look for? (Hint: Cassini wasn't even designed to find the ocean — it discovered the geysers by accident.)

Misconception: "If Enceladus has water, it must have life."

Claude's correction: "Water is necessary for life as we know it, but not sufficient. What Cassini found is that Enceladus has liquid water, organic chemicals, and a heat source — all the *ingredients* that life uses on Earth. But having ingredients doesn't mean the recipe is finished. Scientists call Enceladus a *candidate for habitability* — a place where life might be possible. Whether life actually exists there is something we don't know yet, and answering it would require a future mission to collect and analyze samples directly from the plumes."

Source: Waite et al., Science, 2017 (confirmed in Part B Cassini release).

8 Execution Summary

Metric	Value
Activation Profile	Squadron (12 roles)
Est. Token Budget (median mission)	~255K
Est. Token Budget (flagship, 20+ concepts)	~400K
Model Allocation	Opus 30% / Sonnet 60% / Haiku 10%
Context Windows per Run	5–7
Wave Structure	0, 1A/1B (parallel), 2 (Opus), 3, 3B (CAPCOM)
Safety-Critical Gates	6 BLOCK tests
Pedagogical Quality Tests	8 REVISION gates
Output per Run	Study guides (3 tiers × N concepts) + Try Sessions + Problem sets + Discussion questions + Claude teaching scripts + Assessments + CK YAML entries
Release Version	`v1.50.{MISSION_SLUG}.EDU`
Output Path	`.planning/outputs/missions/{MISSION_SLUG}/edu/`
CK Department Target	Space Science and cross-department links
Parallelism	Independent per mission; run after Part B completes
Resumability	Tracked in `part-c-progress.json`

"The best moment in any science is the moment a student looks at data and says: that's strange. That's not what I expected. What's going on?"

The same thing happened aboard Cassini when the INMS instrument smelled water vapor where models predicted vacuum. The same thing happened to Voyager's navigation team when they saw a volcanic eruption on Io in a photograph taken for an unrelated purpose. The same thing happens in every classroom where a student gets an unexpected result from a Try Session and starts asking the right question.

Part C exists to manufacture that moment — deliberately, at scale, for every mission in NASA's 68-year archive. Not to deliver the answer, but to hand the student the data and ask: *what do you notice?* Claude is there to hear the answer, and help them go further.

Teach the noticing. The science follows.